Slide 3: An unfilled protein sensing niche

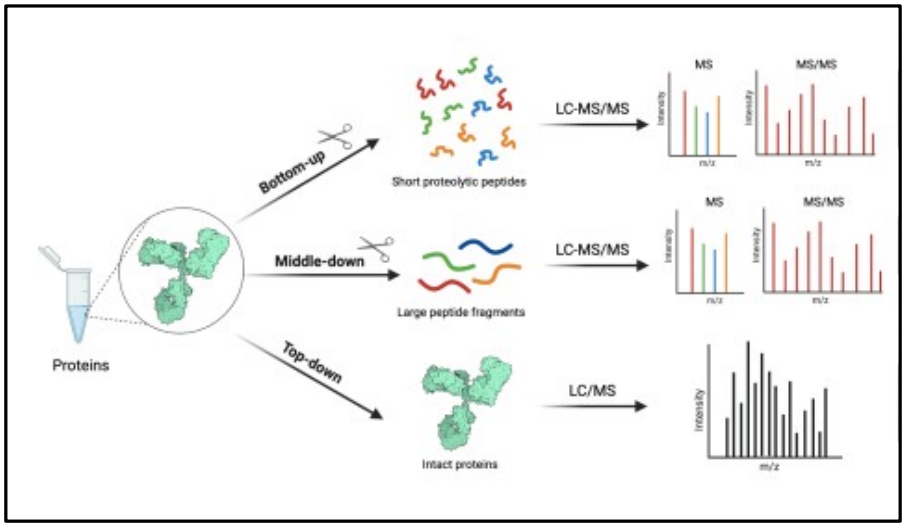


Figure 1: A depiction of a few mass spectrometry techniques in use in proteomics. (Gerpe, 2021) [1].

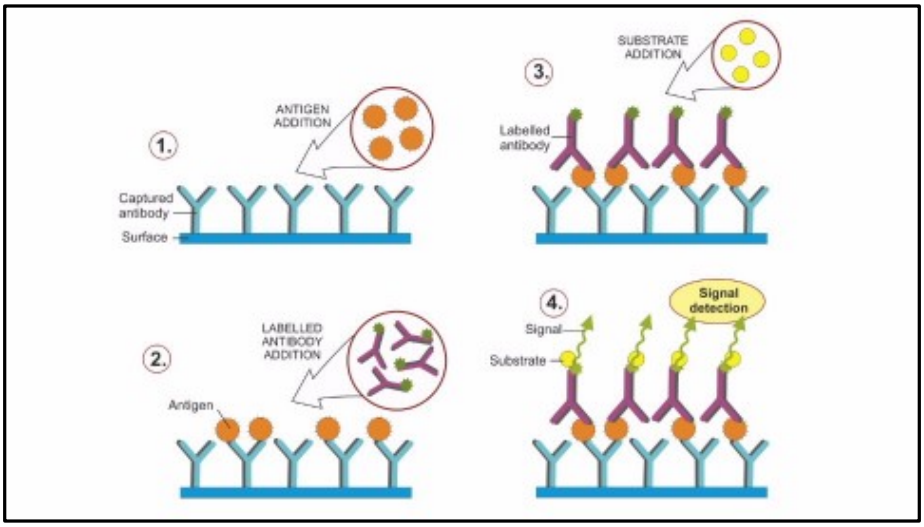


Figure 2: A depiction of the ELISA process. (Agilent Technologies, 2026) [2].

Slide 5: A modular de-novo nanopore protein sensing platform



Figure 3: De-novo protein design of nanopore proteins and mini-binder proteins, which can act as synthetic antibodies (Watson et al., 2023)[3].

Slide 8: Nanopore Surface Coatings

Table 1: Comparison of common nanopore surface modification strategies. Adapted from Eggenberger *et al.*[4] and modified for the design requirements considered in this dissertation.

Method	Phase	Surface bonding	Positive at neutral pH	Diameter control	Organic functionality
Fluid lipid coatings[5–7]	L/C	Noncovalent	Y	N	N
Surfactants[8;9]	L/C	Noncovalent	Y	N	Y^-
Polyelectrolyte coatings[10–14]	L	Noncovalent	Y^+	Y	Y^-
Self-assembled monolayers[15–20]	L/V	Covalent*	Y^-	N	Y
ALD	V	Covalent	Y^-	Y^+	N
MLD	V	Covalent	N	Y^+	Y

Phase: L = liquid phase, V = vapor phase, C = colloid.
Scoring: $\mathbf{Y}^+$: excellent or plentiful; $\mathbf{Y}$: at least some suitable options; $\mathbf{Y}^-$: difficult, uncommon, or dubiously suitable; **N**: nominally impossible or unsuitable for hybrid nanopore formation
*Acid-base condensation (as in silanization) is considered a covalent bond, but thiol-on-gold SAMs may be considered noncovalent depending on the environment.

Slide 11: Polyurea



Figure 4: PU_liner.jpg
An image of a truckbed being lined with PU truck bed liner Ref. (Bullet Liner, 2026) [21]

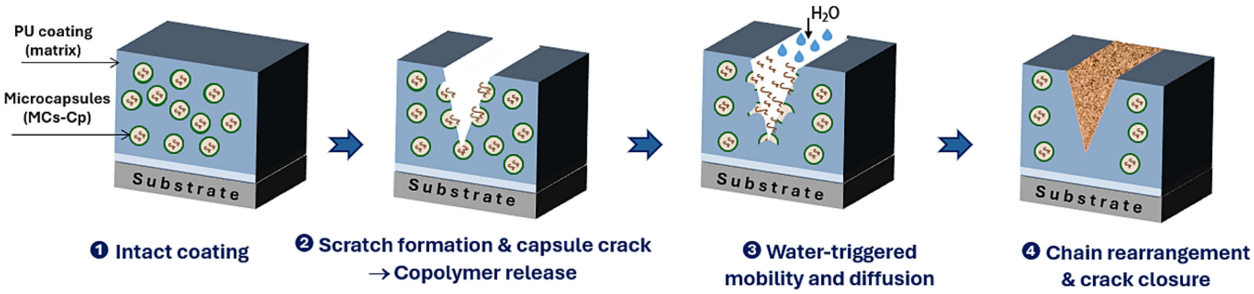


Figure 5: self healing polyurea.jpg
Self healing plastic achieved through the micro-encapsulation of polyruea precursors which react with moisture when exposed by a scratch or fracture Ref. (Avdeliodi et al., 2026) [22]

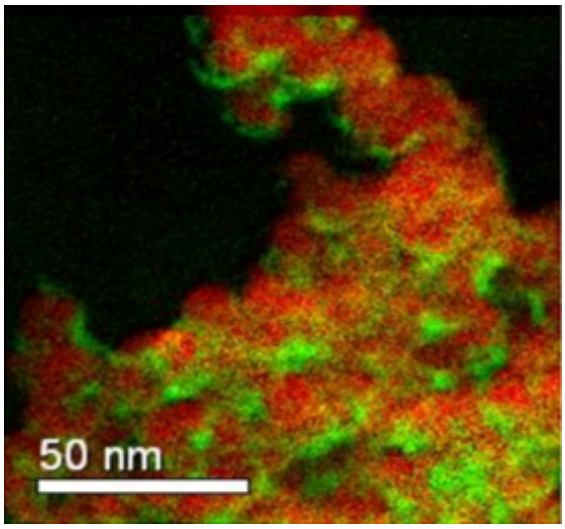


Figure 6: PU_silica_coating.png
A STEM-EDS elemental mapping image of PU coated silica particles Ref. (Mahtabani et al., 2023) [23]

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